What a reward model can show depends on who stopped the episode: failure judges and monitors on a low-cost arm

Ravindu Somawansa
Shisso Robotics, Paris
ravindu.somawansa@shissorobotics.com*

Abstract

Episode length is the failure detector that zero-shot reward models must beat, and how hard it is to beat depends on who stopped the episode. On a public SO-101 benchmark that anyone can rerun, where runs end at a time limit, length alone separates failure from success at AUROC 0.942 within policy. It does not need the timeout. Watched only up to half of the longest episode, it still gives 0.851, because failed runs are slow rather than cut off. Both reward models we test, run zero-shot, score below it, the better one at 0.821. Our own rollouts, which an operator stopped by judgment, reverse the picture: length falls to 0.56, and what catches failures is a judge that sees the finished episode, passing its threshold at an operating point chosen in-sample. No mid-episode monitor reaches its bar. So report a within-policy length baseline, and name the dataset's stopping rule.

1 Introduction

A reward model that scores robot rollouts is also a failure detector, and papers evaluate it as one, mostly against other learned detectors. The baseline that matters is often simpler. When a benchmark stops failed runs at a time limit, the length of an episode encodes its outcome, and so does the identity of the policy that ran (we call the second one the policy-identity null). When an operator stops each run by judgment, both signals collapse and the detection task changes with them. Which of the two a dataset uses is its *stopping rule*, and none of the reward-model papers in our library reports it or the length baseline it implies.

We started with a narrower claim, that elapsed time is the reward model to beat. Our own rollouts broke it. On them length is near chance while a robot-trained judge passed its pre-registered threshold, and since the stopping rule is what separates the two results, it became the subject of the paper.

We measured length on a public SO-101 benchmark [12] and on our own rollouts, with the length analyses frozen before any number existed, and set against it two zero-shot reward models [5, 1] and every judge and monitor we had run on our 88 operator-stopped rollouts.

2 Data and labels

ArmnetBench v0.1 [12] covers the SO-101 with 8 tasks and 7 policies. We use its 2,047 policy rollouts with a success or failure class (515 successes, 1,532 failures) and leave out the teleoperated episodes. One task, pushing a Displayport cable (270 rollouts), has rollouts of a single class and so no per-task AUROC. Our own data are 88 labelled bowls-task episodes on one SO-101 arm with a Jetson Orin NX, and the 50-episode subset (42 successes, 8 failures) that every monitor gate used. The author was the operator: he stopped each run and labelled it at rest, not blind to the policy.

3 The length baseline

On ArmnetBench, episode length is a strong failure detector, and not only because it tells the policies apart (Table 1, Fig. 1). Pooled over policies it reaches 0.938. Within each policy it ranges from 0.91 to 0.96, or 0.942 pair-weighted, with a CI of 0.927–0.956.

Both reward models, run zero-shot, fall below it on this benchmark. Robometer-4B [5], as its authors ship it, scores 0.818 pooled and 0.821 within policy (pair-weighted over its per-policy scores, whose range stands in for a CI; §7 says why). TOPReward [1], with Qwen3-VL-8B [11] and no training, scores 0.661 within policy, and its paired difference to length has a 95 % CI of -0.31 to -0.25. The protocols fixed each model's primary score before any score existed (Robometer's success probability, and the success score of TOPReward's paper). The two descriptive rows of Table 1, read from the same runs, score higher but still below length.

*Code, result files and the checks that regenerate every number: `github.com/ravediamond/paper-reward-models-stopping-rule`. The frozen protocols are copied verbatim under `repro/`; the bar commits named in the paper belong to the authors' research repository `ravediamond/road-to-robotics`.

signal	pooled	within policy
episode length	0.938 (0.925–0.951)	0.942 (0.927–0.956)
length seen up to 50 % of the longest run	—	0.851 (0.829–0.873)
Robometer-4B, success probability	0.818 (0.790–0.843)	0.821 (per-policy 0.79–0.87)
Robometer-4B, last progress (descriptive)	0.838 (no CI)	—
TOPReward, success score	0.653 (0.623–0.684)	0.661 (0.631–0.692)
TOPReward, Yes/No component (descriptive)	—	0.678

Table 1: ArmnetBench, failure AUROC with 95 % episode-bootstrap CIs. Read down the within-policy column: no reward-model score reaches length.

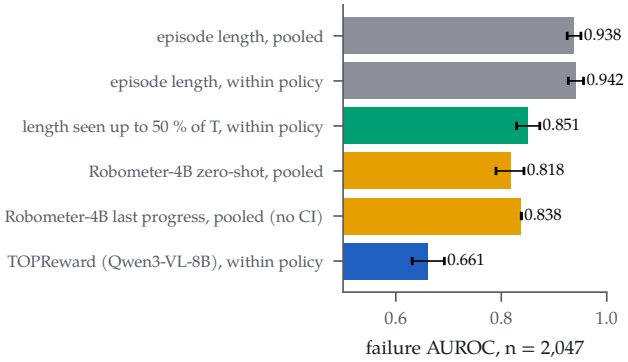


Figure 1: Failure AUROC on ArmnetBench: episode length against two zero-shot reward models. Length leads both, pooled and within policy (95 % CIs where available).

4 Length is more than the timeout

If length only reflected the timeout, its signal would sit in the few runs that hit it. It does not. Taking each task's longest episode as its timeout, only 8.7 % of failures run that long. At half that length 88 % of failures are still running but only 19 % of successes, and length observed up to that cut gives 0.851 within policy (CI 0.829–0.873; Fig. 2). TOPReward on frames up to the same cut reaches 0.667. Failed rollouts are slow, not timed out, and both readings fixed in the protocol hold.

On our own operator-stopped rollouts the same baseline is at chance: 0.56 on the 69 rollouts TOPReward scored and 0.52 on all 88. The baseline that matters here is the policy-identity null, because 17 of the 25 failures in the 69 come from one policy, TurboVLA [16]. Neither reward model beats it. TOPReward scores 0.82 (CI 0.70–0.93) against 0.87 for policy identity alone, and 0.78 on the 50 episodes without TurboVLA. Robometer scores 0.577 against 0.878 on all 88.

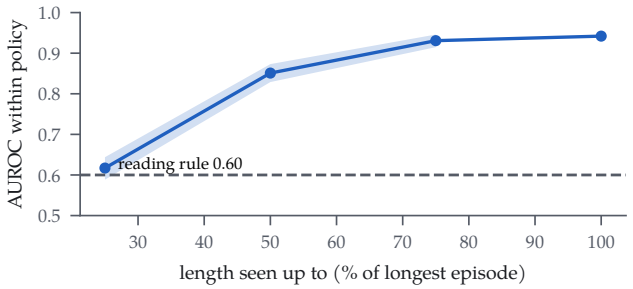


Figure 2: Length observed up to a fraction of the task's longest episode, AUROC within policy: most of the signal is there by half the episode (band: 95 % CI).

5 Judges and monitors on our rollouts

Table 2 gives each judge and monitor a threshold-free AUROC where one is defined, then failures caught and successes flagged at its operating point, which for the judges at rest is chosen in-sample.

A judge that sees the finished episode works on these rollouts, and the input format matters more than the robot fine-tune. A deterministic top-camera check (colour masks and geometry, no learning) is right on 94.3 % of judged episodes, with 0 false positives (Wilson 0–13 64-frame two-view clip and NVIDIA's prompt, Flora-Rewarder-27B [8] reaches a threshold-free AUROC of 1.00 (bootstrap CI 1.00–1.00). At its in-sample operating point it flags 8 of 8 failures at 2 of 42 successes (Wilson on the false alarms 1–16 bar of at least 6 of 8 failures at no more than 2 of 42 successes. Its base model Qwen3.5-27B [10], without the robot fine-tune, passes the same bar on the same input: AUROC 0.91 (0.78–1.00), 7 of 8 at 2 of 42. The smaller Qwen3.5-9B [10] and Gemini 3.8 Flash [2] fail it, the 9B by flagging 16 successes and Gemini by one false alarm too many.

On these 50 rollouts, no mid-episode monitor reaches its bar of at least 5 of 8 failures at no more than 1 of 20 false interventions (Fig. 3). The surprise of a separate world model, trained in the LeWorldModel recipe [7], comes closest with 2 of 8 at 2 of 42 (AUROC 0.86). A Gemini sliding-window judge catches 0. The action-conditioned risk head on the world model's latent has a history. Its first version reached AUROC 0.93, but it read elapsed time, so we withdrew that result and pre-registered a second version without time or length. The second version catches 3 of 8 at 5 of 42, with AUROC 0.79 against 0.71 when its actions are shuffled, and its MLP variant catches 6 of 8 but flags 13 of 42.

6 Related work

Robometer [5] and TOPReward [1] are the two reward models we test. Robometer scales a general-purpose reward model through trajectory comparisons, TOPRe-

method	kind	AUROC	caught / 8	flagged / 42	threshold
colour judge (no learning)	at rest	— (94.3 % right, 0 FP, 5 FN)	—	—	fixed rule
Flora-Rewarder-27B	at rest	1.00 (1.00–1.00)	8	2	in-sample
Qwen3.5-27B, same input	at rest	0.91 (0.78–1.00)	7	2	in-sample
Qwen3.5-9B, same input	at rest	0.75	7	16	in-sample
Gemini 3.8 Flash, same input	at rest	0.96	8	3	in-sample
Gemini sliding windows	monitor	—	0	1	fixed before scoring
world-model surprise	monitor	0.86	2	2	fixed before scoring
world-model latent drift	monitor	0.30	0	2	fixed before scoring
risk head, logistic (v2)	monitor	0.79	3	5	5-fold by episode
risk head, MLP (v2)	monitor	—	6	13	5-fold by episode

Table 2: Judges and monitors on the 50 operator-stopped rollouts (42 successes, 8 failures). Judges see the finished episode; monitors are held to $\geq 5/8$ caught at $\leq 1/20$ false interventions.

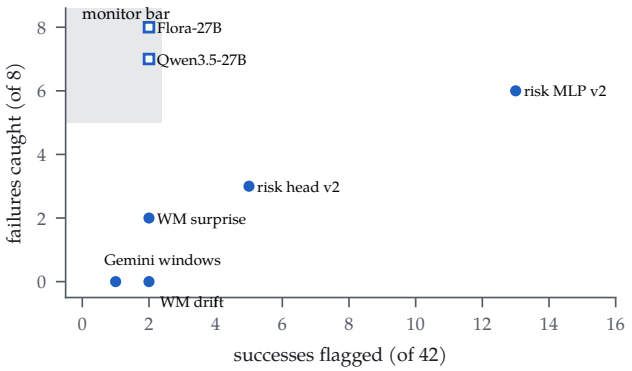


Figure 3: Failures caught against successes flagged on the 50 rollouts. No monitor (circles) reaches its shaded bar; the judges at rest (squares) sit at their in-sample operating point, where the monitor bar does not apply.

ward reads the token probabilities of a video-language model as hidden rewards, while a survey of verifiers for robot policies [14] maps the wider family. Our monitors borrow from the mid-episode line of work. FAIL-Detect [17] and SAFE [3] learn uncertainty- and feature-based failure scores for imitation and VLA policies. FARM [9] reads failure signals from the internal states of a frozen world model, close to our world-model monitors, and ReAP [6] predicts action-conditioned risk, the idea behind our risk head. VLA-Corrector [13], Spotter [4] and REMORA [15] add stage verification, a reflecting VLM and recurrent progress tracking. ArmnetBench [12] and a standardized SO-101 VLA benchmark [18] evaluate policies on the same low-cost arm. None of these works reports the elapsed-length baseline under the dataset's stopping rule, and that comparison is what this paper adds.

7 Limitations

The judge pass on our own rollouts rests on 8 failures and on operating thresholds chosen on the same 50 episodes, labelled by a single operator, the author, who was not blind to the policy. The threshold-free AUROCs do not depend on that choice, and fresh rollouts scored at the frozen thresholds are the confirming test. Policy identity is a second confound: 17 of the 25 failures in our 69 rollouts are TurboVLA's, so any score that reads the policy gets the policy-identity null for free. The risk head is trained and tested on the same 50 episodes (five folds by episode keep episodes from leaking, but tasks and policies do leak). Robometer's within-policy number has no episode-bootstrap CI and Robometer is missing from the truncation analysis, because its per-episode scores stayed on an older machine; a copy from that machine or a rescore of about $6 would restore both. TOPReward's paper does not give its prefix count, frames per query or Yes/No wording, so those are our choices. The ArmnetBench result covers eight tasks, and the timeout is approximated by each task's longest episode.

8 Conclusion

Before evaluating a reward model as a failure detector, run episode length within policy on the same rows, and say how the episodes ended. On a benchmark that stops at a time limit, length was the detector to beat, and neither reward model beat it. On rollouts an operator stopped, length was useless, and what caught failures was a judge that saw the whole finished episode in the right input format.

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analysis	repro folder	expected files	protocol
length baseline, truncation	repro/length-null	length_armnetbench.json, counts_ci.json	LENGTH_PROTOCOL.md, frozen `1a69f30`
Robometer on ArmnetBench	repro/robometer-judge	g9a_ab_complete.json	PROTOCOL.md (primary score fixed there)
Robometer and length on own rollouts	repro/robometer-judge	g9a_own.json	PROTOCOL.md
colour judge	repro/robometer-judge	g9c3_test_results.json	PROTOCOL.md
TOPReward, both datasets	repro/topreward	topreward.json, own_rollout_labels.csv	TOPREWARD_PROTOCOL.md, `6241570`
judges at rest, input-format gate	repro/flora-input-format	g_flora_summary.json, g_flora_abl_summary.json	PROTOCOL.md
window judge, world-model monitors	repro/six-monitors	judge_tau.json, w1a_summary.json	PROTOCOL.md
action-conditioned risk head	repro/reap	monitors.json, monitors_v2.json	PROTOCOL.md

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Appendix: from plain name to analysis

Every number is recomputable from the repository; this table maps each analysis to its files.